SIKSHA 'O' ANUSANDHAN

(Deemed to be University)

Faculty of Engineering & Technology (ITER)

Department of Computer Science and Engineering

<u>Project Proposal Form</u>

FINAL YEAR RESEARCH PROJECT-2026

SECTION: 2241044 **GROUP NO: 51-09**

PROJECT TITLE: "Development of an Integrated Hybrid Optimization TinyML Framework for Real-Time Environmental Monitoring on Microcontroller."

PROJECT ABSTRACT:

This research evaluates an integrated hybrid optimization approach to develop a high performing "Teacher" model which is then shrunk into a smaller "Student" version that fits with microcontroller's limited memory. By training this model specifically on temperature, humidity, and smoke patterns, we enable the device to monitor the surroundings. We can then simulate the model on an online microcontroller to prove it can provide life-saving alerts with minimal delay and low power consumption.

The core problem is the incompatibility between deep learning models and embedded hardware where models require high memory and computational power which makes them unsuitable for deployment on devices with limited RAM and energy. Existing studies treat optimization techniques in isolation, resulting in incomplete understanding and inefficient developments. Due to clear research gaps in how these techniques are applied and evaluated, these is a strong need for an integrated and experimentally validated framework which can monitor the environment and can provide lifesaving alerts which can be processed locally on a microcontroller.

The main role of the system is to transform a complex neural network into a smaller, executable binary that can perform real-time environmental monitoring locally on a device without constant need of cloud server. The technical core involves a Teacher-Student model where a model learns from a larger baseline through Knowledge Distillation and to reduce the memory and energy usage, we have implemented iterative pruning to remove redundant parameters and Quantization-Aware Training to convert weights into 8-bit integers. The final model is simulated in an online microcontroller environment, where we measure and benchmark the RAM usage, interface latency and accuracy retention. This ensures the model remains high performing while fitting within hardware memory restrictions.

This research enables efficient on-device intelligence, reducing reliability on cloud, lowers energy consumption, and improves data privacy. This has applications in affordable industrial monitoring, environmental sensing, and remote healthcare systems. This project contributes toward making advanced AI accessible, scalable and deployable in real-world environments.

(1) SOFTWARE, HARDWARE OR METHODS/ALGORITHMS SPECIFICATIONS:

Software Specifications:

1. **Development Environment: Google Colab (Python-based)**- Used for high-speed model training without requiring local GPU resources.
2. **Machine Learning Framework: TensorFlow 2.x and Keras**-Used for building the baseline CNN/RNN architectures.
3. **Optimization Libraries: TensorFlow Model Optimization Toolkit (tfmot)**- Used to implement the specific pruning and quantization-aware training (QAT) algorithms.
4. **Edge Conversion: TensorFlow Lite Converter-** Used to transform heavy Python models into .tflite flatbuffers for embedded execution.
5. **Simulation & Testing: Wokwi Simulator**- Used to emulate the microcontroller hardware and measure execution speed and memory usage.

"Virtual" Hardware Specifications:

1. **Target Microcontroller: ESP32 (Dual-core, 520KB SRAM)**- This is our primary target because it is widely used in real-world IoT.
2. **Instruction Set Architecture: Xtensa® Dual-Core 32-bit LX6**- It allows us to test how our optimizations perform on standard embedded CPUs.
3. **Virtual Sensors: DHT11 (Temp/Humidity) & MQ-series (Smoke/Gas)** – These will be emulated in the Wokwi environment to provide the raw input.

"Physical" Hardware Specifications:

1. **Standard Laptop or Desktop**-This will be a Development Workstation.
2. **Internet Connection**- Required to access Google Colab and Wokwi.

Methods/Algorithms Specifications:

1. **Architecture**- **CNN** for pattern-based anomalies or **LSTM/RNN** for time-series sensor data.
2. **Knowledge Distillation Algorithm**- Uses a Dual-Loss Function involving Cross-Entropy and Kullback-Leibler (KL) Divergence to transfer intelligence from a "Teacher" to a "Student".
3. **Pruning Algorithm**- Iterative Magnitude-based Pruning, which uses a binary mask to zero-out weights that contribute the least to the final output.
4. **Quantization Algorithm**- 8-bit Linear Quantization, converting 32-bit floating-point weights into 8-bit integers to reduce memory footprint.

(2) NAME, REG. NO AND SIGNATURE OF GROUP MEMBERS:

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(3) APPROVAL STATUS:

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Project Supervisor	**Section Coordinator, FRP**	**FRP Coordinator**